

LAB EXPERIMENT

ITA0506-COMPUTER VISION

G Chandra Sekhar Reddy(192472173)

EXPNO:22

Aim

To sharpen an image using a Laplacian mask with a positive center coefficient.

Laplacian Mask

A commonly used 4-neighbor Laplacian mask with a positive center coefficient is:

$$\begin{bmatrix} 0 & -1 & 0 \\ -1 & 4 & -1 \\ 0 & -1 & 0 \end{bmatrix}$$

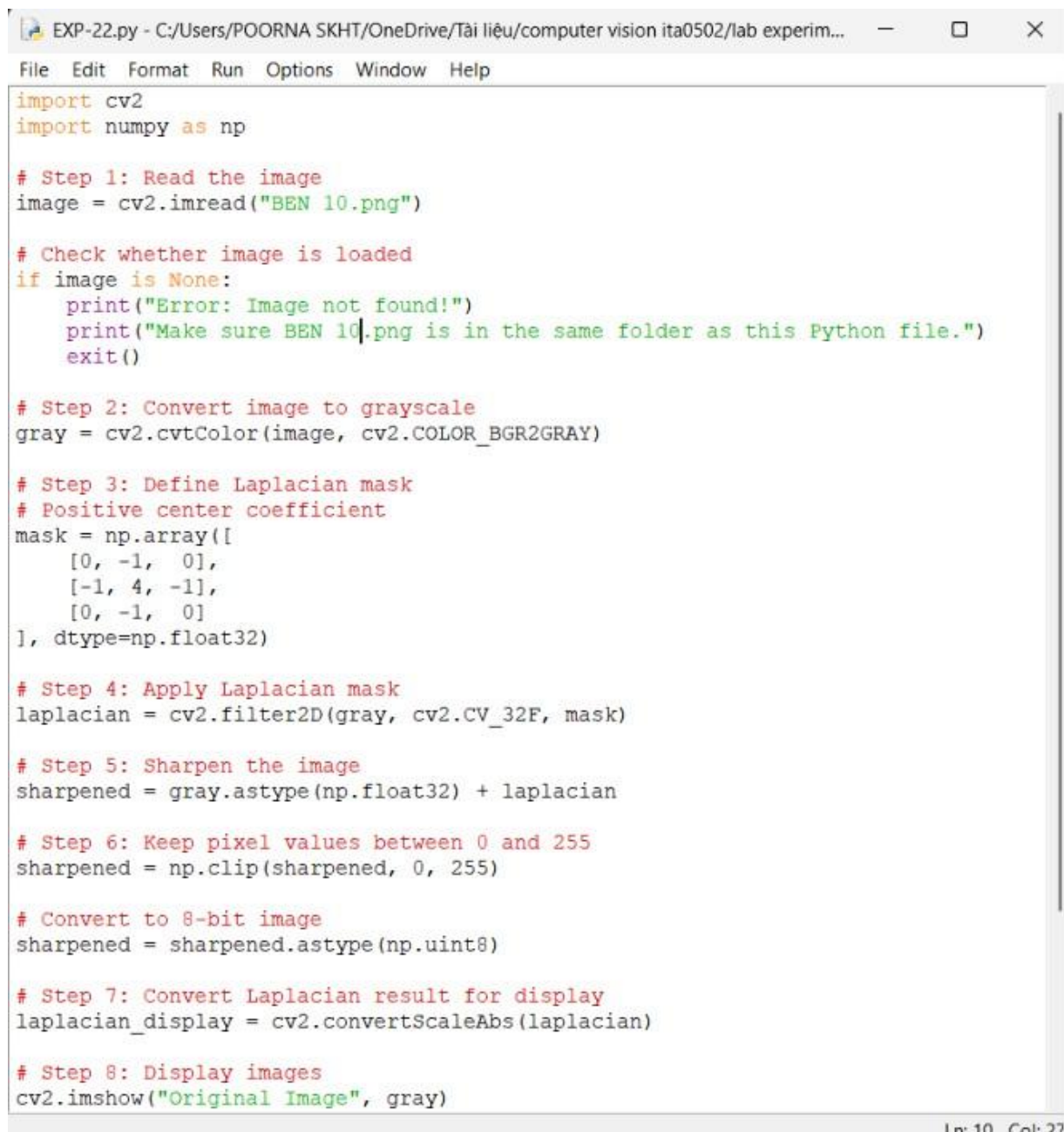
For sharpening, the Laplacian result is added to the original image:

Sharpened Image = Original Image + Laplacian Image

Algorithm

1. Read the input image.
2. Convert the image to grayscale.
3. Define the Laplacian mask with a positive center coefficient.
4. Apply the mask using convolution.
5. Add the Laplacian image to the original image.
6. Limit pixel values between 0 and 255.
7. Display the original and sharpened images.

Code:

A screenshot of a text editor window titled "EXP-22.py - C:/Users/POORNA SKHT/OneDrive/Tài liệu/computer vision ita0502/lab experim...". The window has a menu bar with "File", "Edit", "Format", "Run", "Options", "Window", and "Help". The code is written in Python and includes several comments in red. The code performs the following steps: 1. Import cv2 and numpy as np. 2. Read the image "BEN 10.png" using cv2.imread(). 3. Check if the image is loaded; if None, print an error message and exit. 4. Convert the image to grayscale using cv2.cvtColor(). 5. Define a Laplacian mask as a 3x3 numpy array. 6. Apply the Laplacian mask using cv2.filter2D(). 7. Sharpen the image by adding the Laplacian result to the grayscale image. 8. Clip the pixel values to the range [0, 255] using np.clip(). 9. Convert the image to 8-bit using astype(np.uint8). 10. Convert the Laplacian result to a scale suitable for display using cv2.convertScaleAbs(). 11. Display the original image using cv2.imshow().

```
import cv2
import numpy as np

# Step 1: Read the image
image = cv2.imread("BEN 10.png")

# Check whether image is loaded
if image is None:
    print("Error: Image not found!")
    print("Make sure BEN 10.png is in the same folder as this Python file.")
    exit()

# Step 2: Convert image to grayscale
gray = cv2.cvtColor(image, cv2.COLOR_BGR2GRAY)

# Step 3: Define Laplacian mask
# Positive center coefficient
mask = np.array([
    [0, -1, 0],
    [-1, 4, -1],
    [0, -1, 0]
], dtype=np.float32)

# Step 4: Apply Laplacian mask
laplacian = cv2.filter2D(gray, cv2.CV_32F, mask)

# Step 5: Sharpen the image
sharpened = gray.astype(np.float32) + laplacian

# Step 6: Keep pixel values between 0 and 255
sharpened = np.clip(sharpened, 0, 255)

# Convert to 8-bit image
sharpened = sharpened.astype(np.uint8)

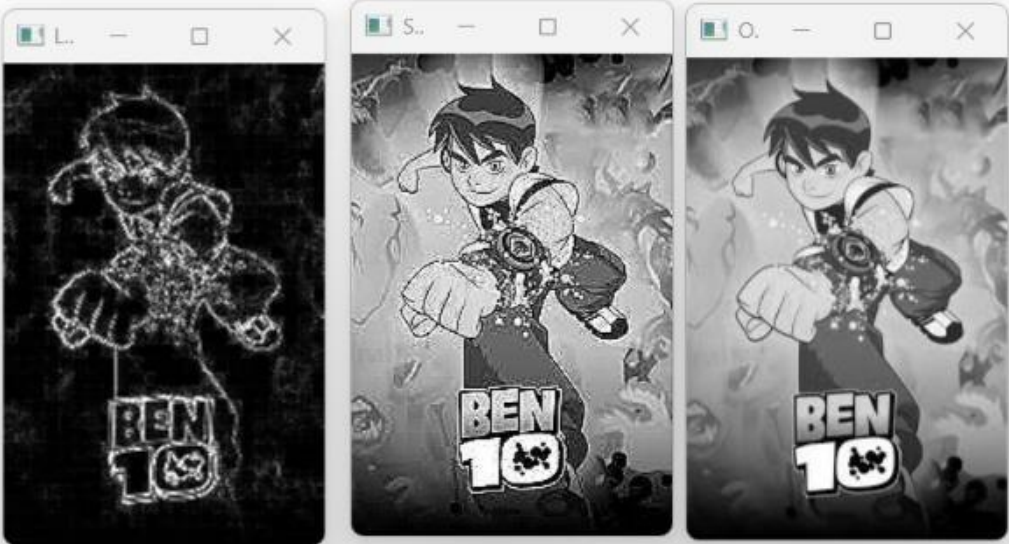
# Step 7: Convert Laplacian result for display
laplacian_display = cv2.convertScaleAbs(laplacian)

# Step 8: Display images
cv2.imshow("Original Image", gray)
```

Ln 10, Col 27

OUTPUT :

```
*IDLE Shell 3.13.3*
File Edit Shell Debug Options Window Help
Python 3.13.3 (tags/v3.13.3:6280bb5, Apr 8 2025, 14:47:33) [MSC v.1943 64 bit (AMD64)] on win32
Enter "help" below or click "Help" above for more information.
>>>
= RESTART: C:/Users/POORNA SKHT/OneDrive/Tài liệu/computer vision ita0502/lab experiments/EXP-21.py
= RESTART: C:/Users/POORNA SKHT/OneDrive/Tài liệu/computer vision ita0502/lab experiments/EXP-22.py
```



Ln: 7 Col: 0

Result

The image is successfully sharpened using a Laplacian mask with a positive center coefficient. The edges and fine details of the image become more prominent.